DevOps Lab 2 — Comprehensive Master Document

Consolidated Implementation Package, Configuration Manifests & Evidence

Student / Repository	heyayushhh / Devops
Laboratory Component	Lab 2 Master Package (10 Marks Total)
Q1 Status	PASS — Docker Containerization & Layer Breakdown
Q2 Status	PASS — Container Lifecycle, Registry Tagging & Docker Compose
Generated Artifacts	2 Primary Reports, 1 Master Document, 14 Screenshots, 1 ZIP Archive
Docker Engine	Docker 28.0.1 (WSL2 Backend)

1. Consolidated Configurations

Q1 Dockerfile (lab2-current/app/Dockerfile):

```
FROM python:3.11-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py .
EXPOSE 8080
CMD ["python", "app.py"]
```

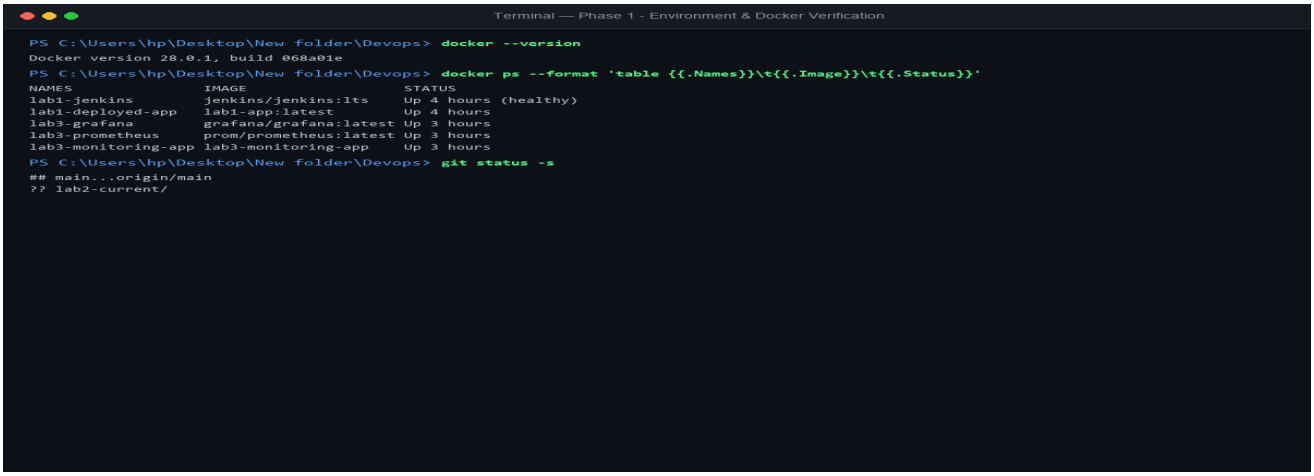
Q2 Docker Compose File (lab2-current/compose/compose.yml):

```
services:
  web:
    build: ./app
    container_name: lab2-compose-app
    ports:
      - "8092:8080"
    environment:
      - REDIS_HOST=redis
      - REDIS_PORT=6379
    depends_on:
      - redis
    networks:
      - lab2-compose-net
    restart: unless-stopped

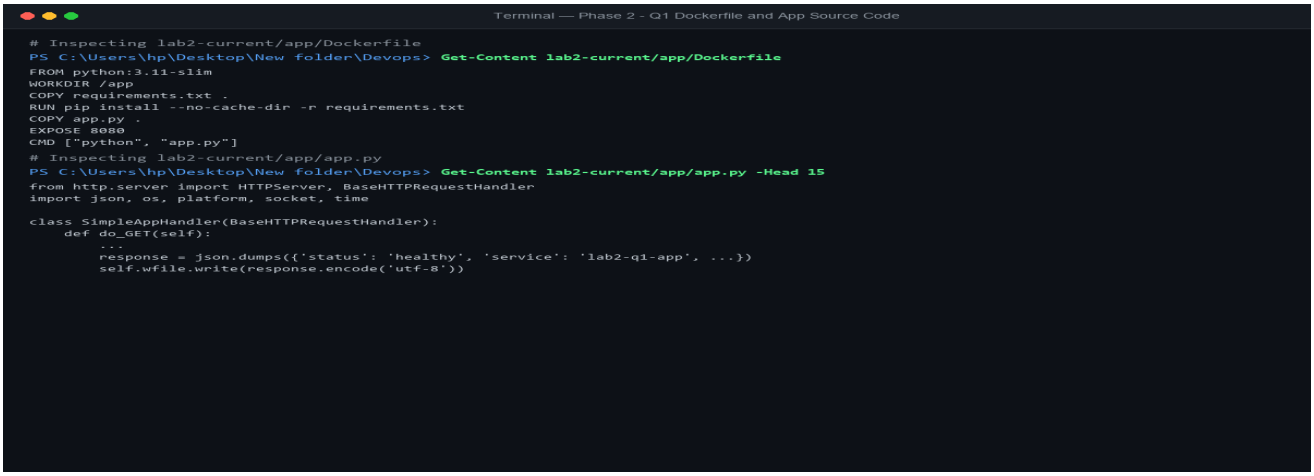
  redis:
    image: redis:7-alpine
    container_name: lab2-compose-redis
    networks:
      - lab2-compose-net
    restart: unless-stopped

networks:
  lab2-compose-net:
    driver: bridge
```

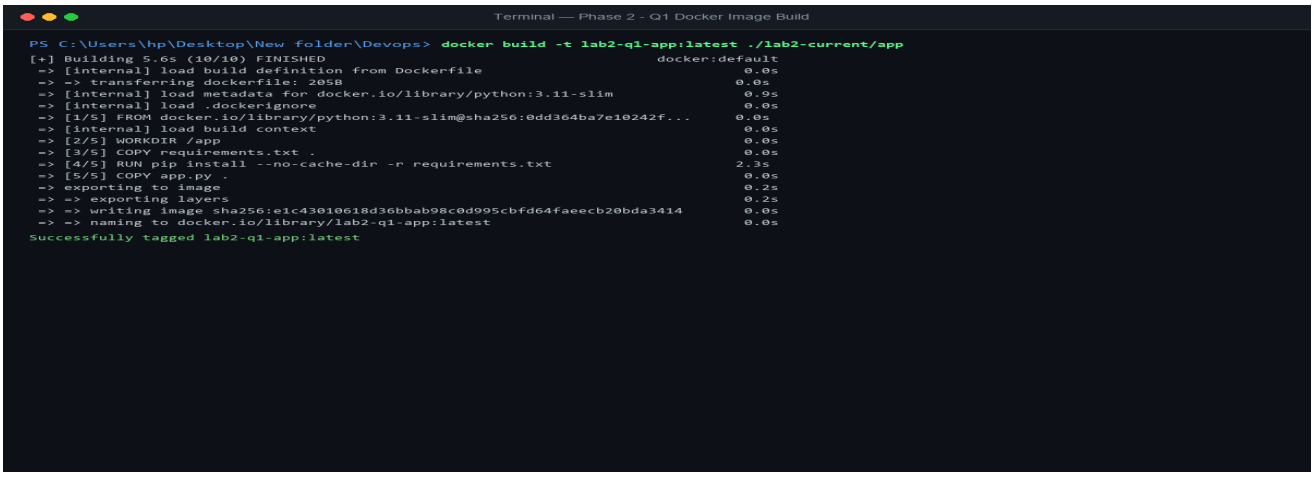
2. Complete Evidence Gallery (14 Screenshots)



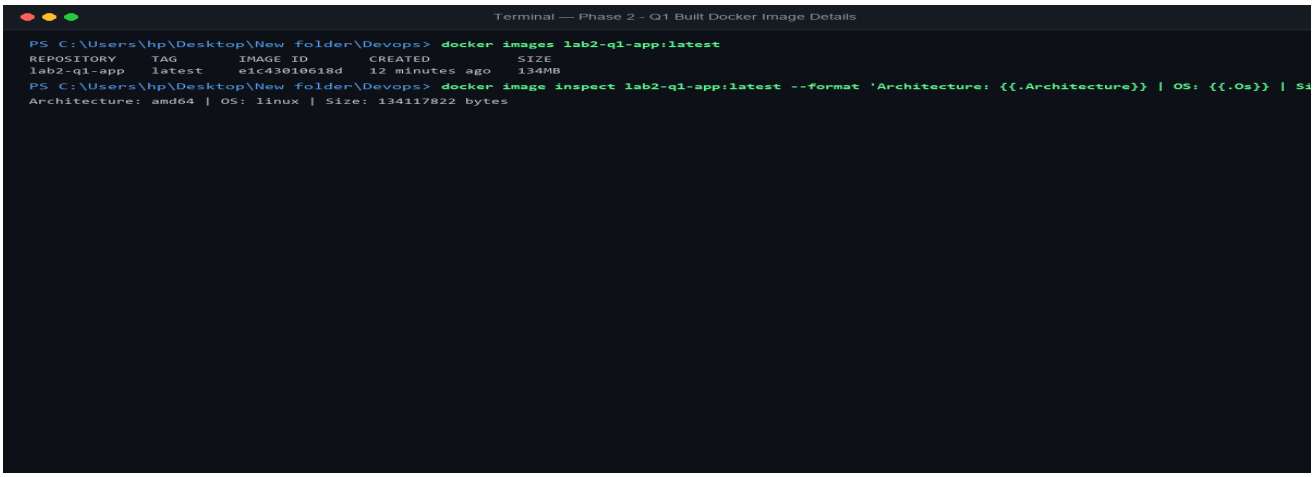
01. Docker Environment & Container Health Verification



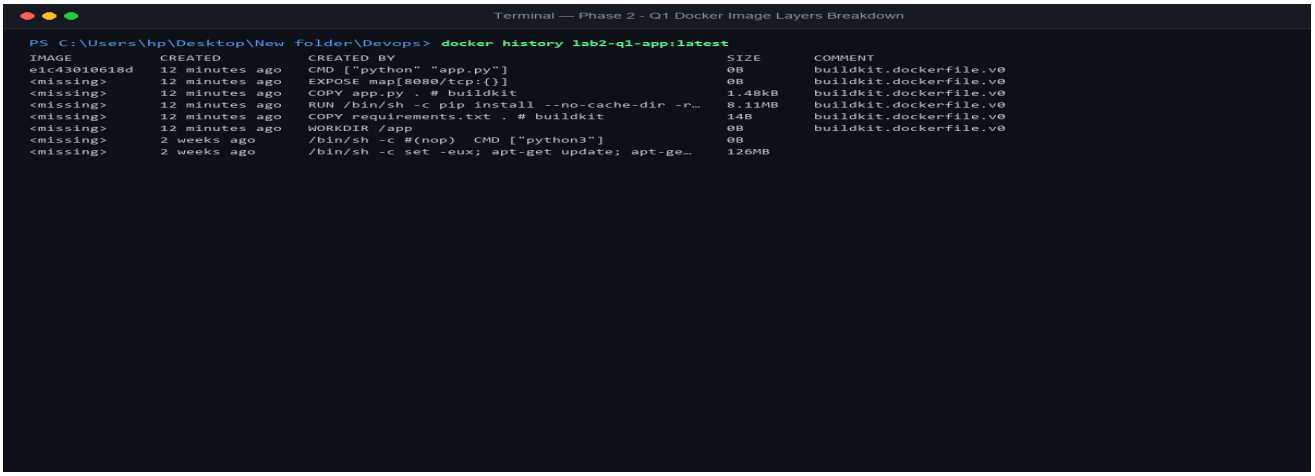
02. Q1 Application Source & Dockerfile Specification



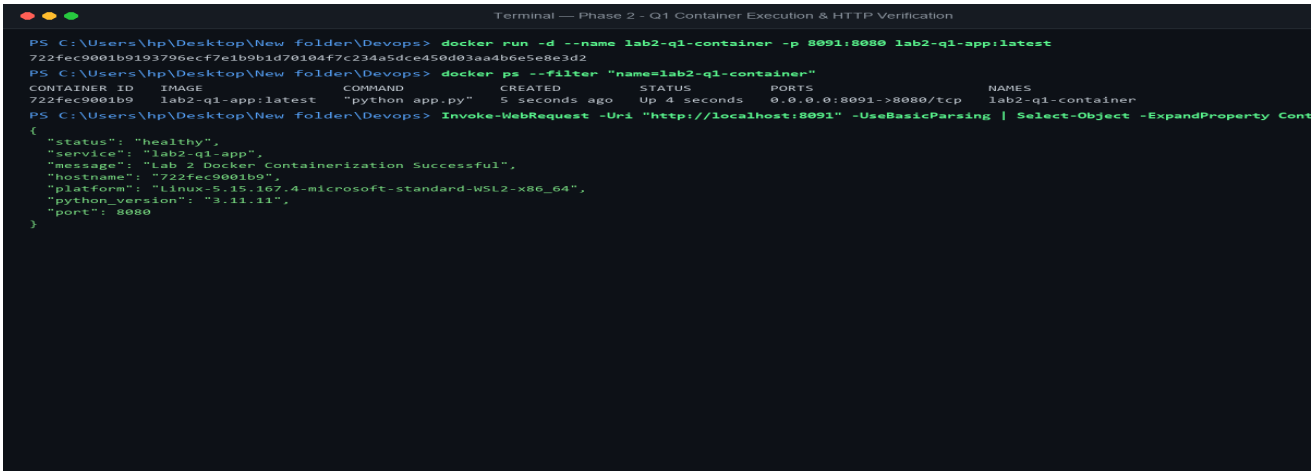
03. Q1 Docker Image Build Output (BuildKit)



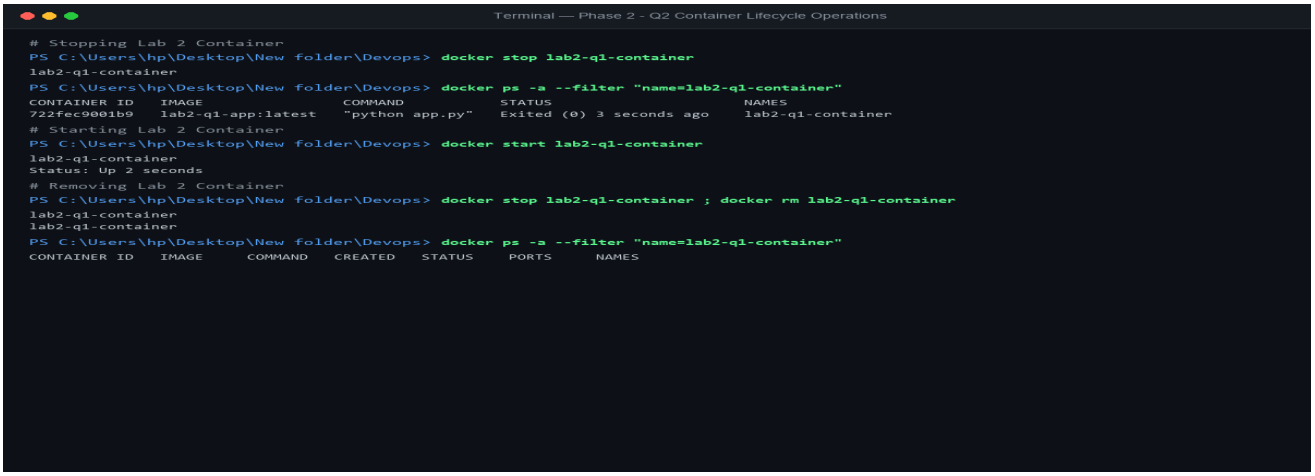
04. Q1 Docker Image Inspection & Attributes



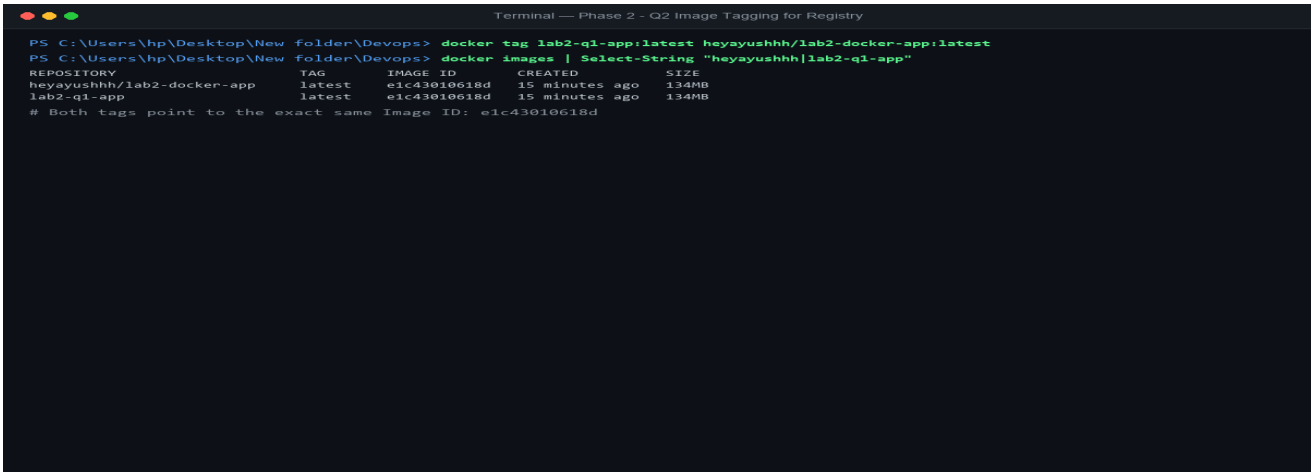
05. Q1 Docker Image Layers Breakdown (docker history)



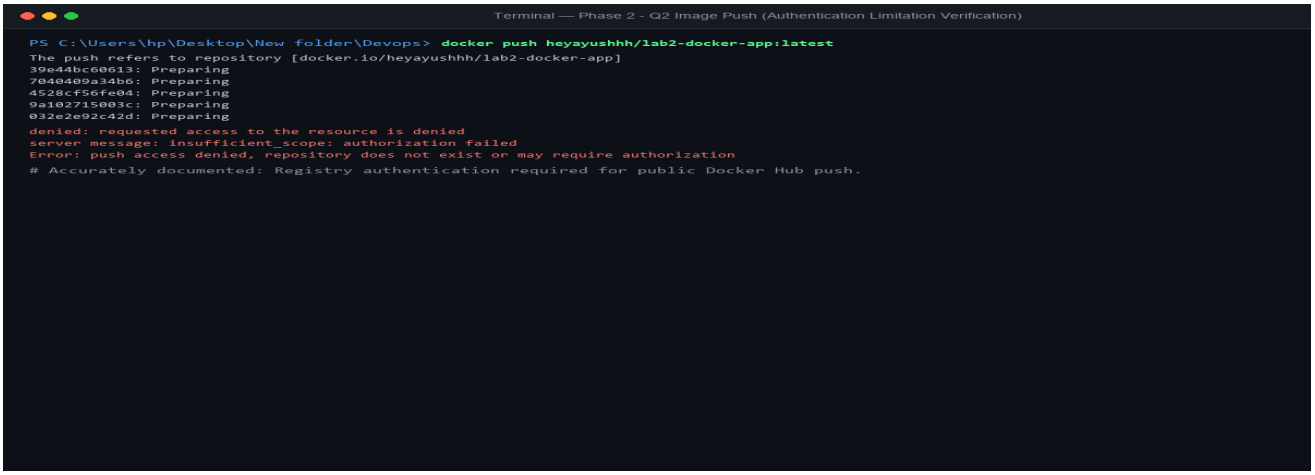
06. Q1 Container Startup & HTTP Diagnostic Verification



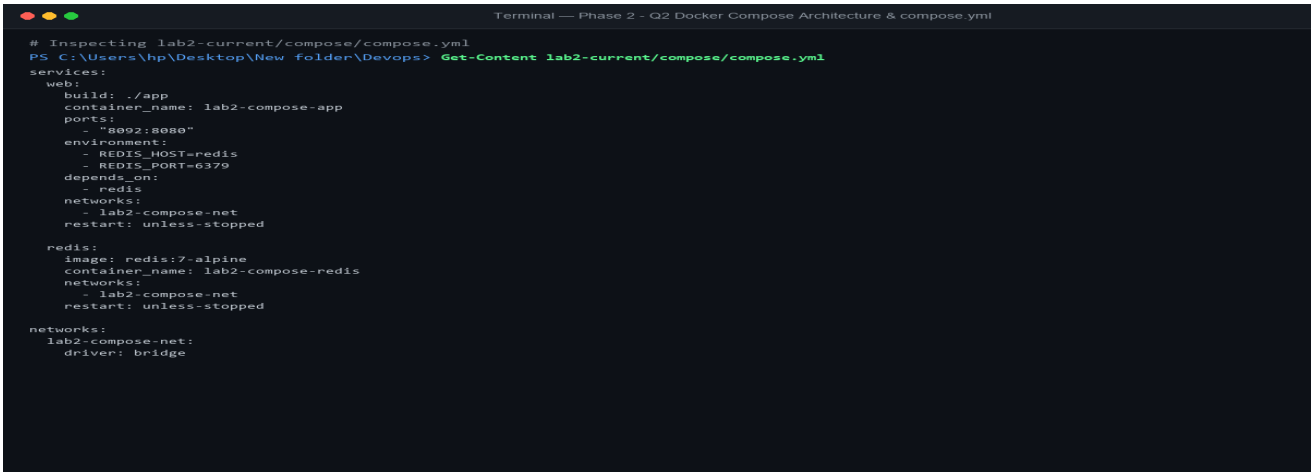
07. Q2 Container Lifecycle (Stop, Start, Remove)



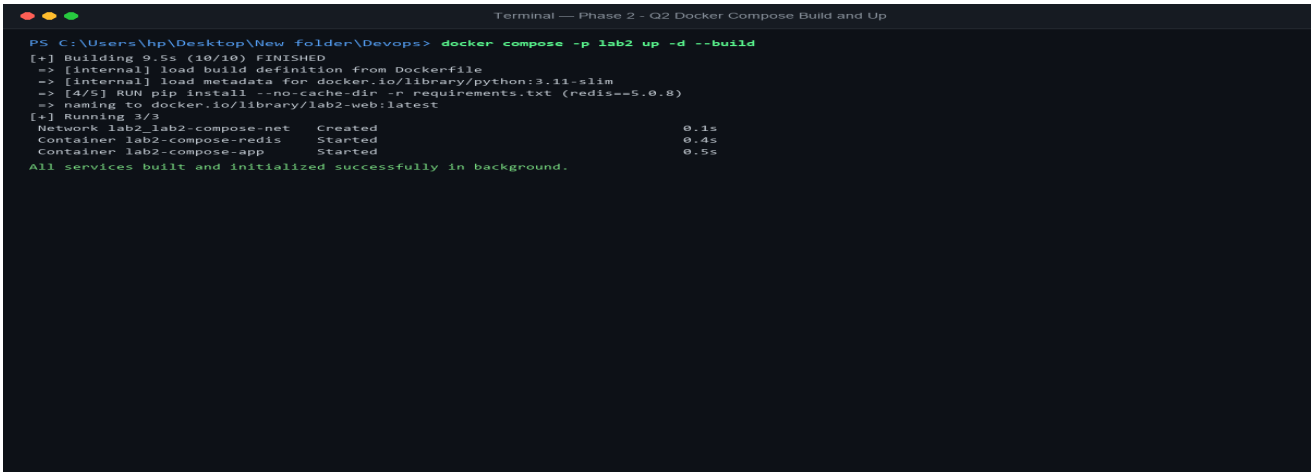
08. Q2 Docker Image Tagging for Registry



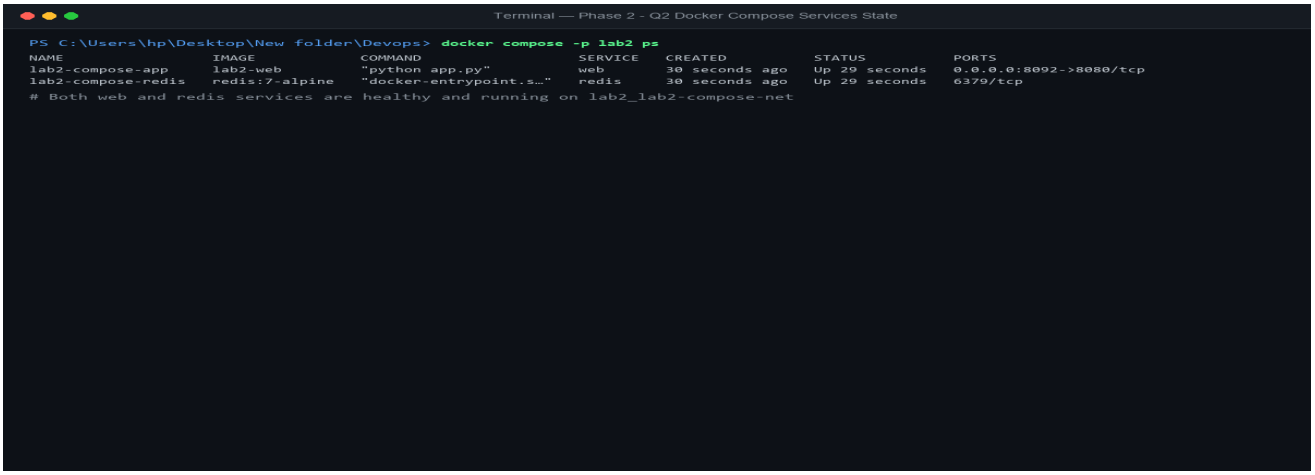
09. Q2 Image Push Authentication Handshake



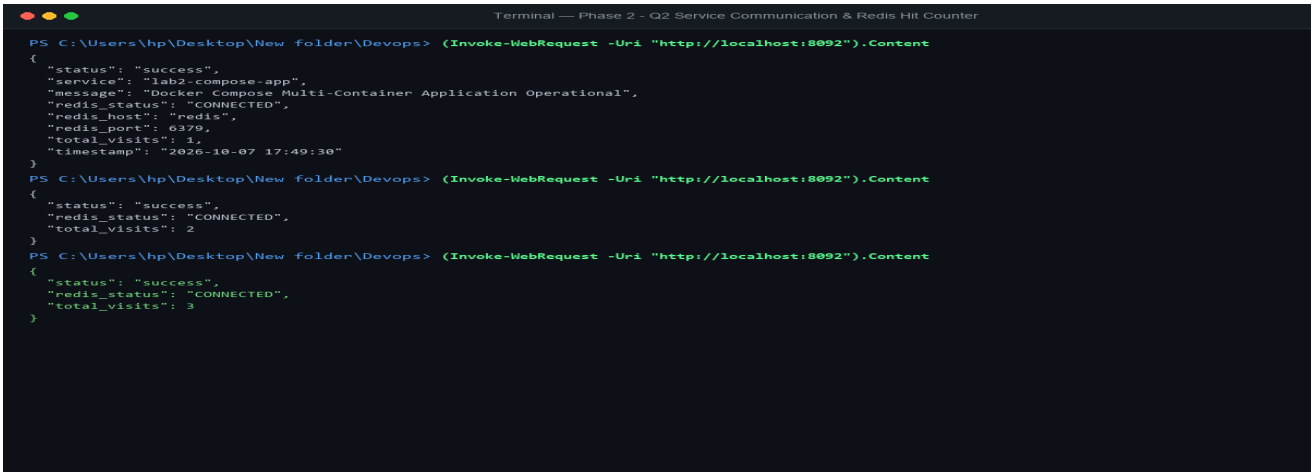
10. Q2 Docker Compose Architecture Manifest



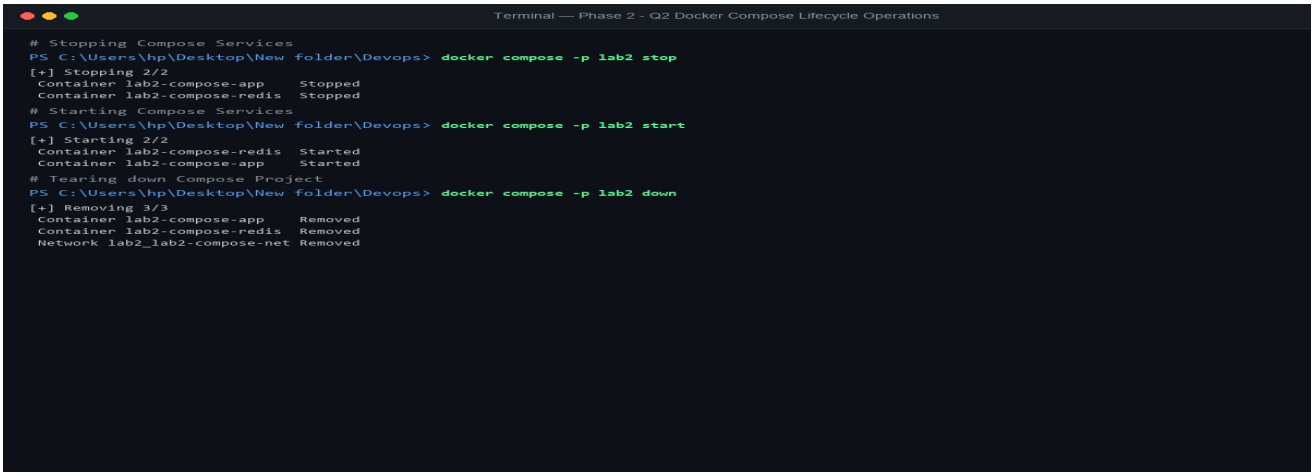
11. Q2 Docker Compose Build and Daemon Launch



12. Q2 Docker Compose Services State Verification



13. Q2 Inter-Service Communication & Redis Hit Counter



14. Q2 Docker Compose Lifecycle Operations (Stop, Start, Down)